

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

September 30, 2026

Outline

- 1 SoPS Lecture 11
 - Königsberg bridge problem
 - Math and creativity
 - Wrapping up

Review

Last time

Review

Last time

- we learned the degree sum formula
- we learned Euler's formula for planar graphs
- questions about Sprouts

First-year/Transfer Event



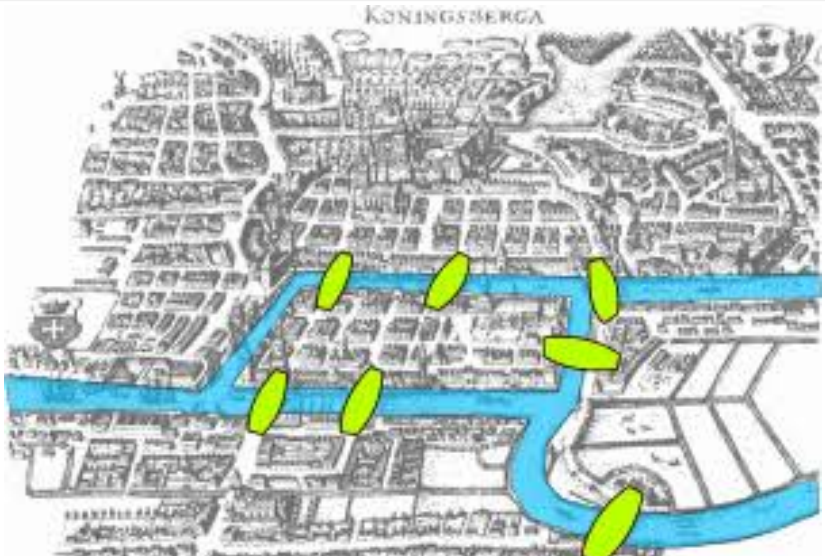
Outline

- 1 SoPS Lecture 11
 - Konigsberg bridge problem
 - Math and creativity
 - Wrapping up

Bridges of Königsberg



Bridges of Königsberg



Bridges of Königsberg

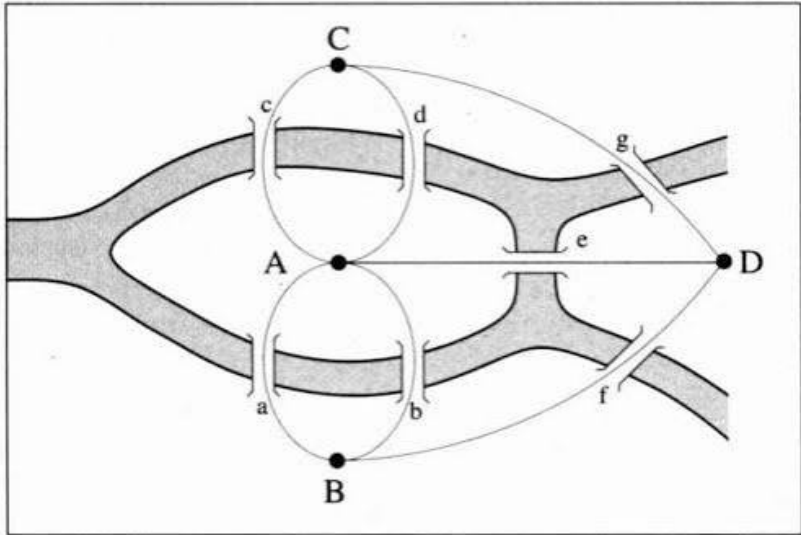
Problem:

Can one find a route which crosses every bridge exactly one time?

Bridges of Königsberg



Bridges of Konigsberg



Eulerian Paths

Defintion

An **Eulerian path** in a graph is a sequence of edges that you can walk along to visit every edge exactly one time.

Eulerian Paths

Eulerian Paths

Theorem:

A graph has an Eulerian path if and only if it has less two or less nodes with odd degree.

Eulerian Paths

Theorem:

A graph has an Eulerian path if and only if it has less two or less nodes with odd degree.

Corollary:

The Konigsberg bridge problem has no solution!

Math is creative

Math is creative

- Euler's Konigsberg bridge problem.

Math is creative

- Euler's Konigsberg bridge problem.
- $1 + 3 + 5 + 7 + \cdots + 2n + 1 =$

Creative Konigsberg solution



Outline

- 1 SoPS Lecture 11
 - Konigsberg bridge problem
 - Math and creativity
 - Wrapping up

Challenge problem

Activity:

Color every dot in a 4×4 grid red or blue, avoiding any rectangle with horizontal and vertical sides whose four corners have the same color. Rectangles of every size count. If you succeed, try 5×5 .

Challenge problem

Discussion:

What did you try and what did it reveal?

Computation vs. Creative Math

Computation vs. Creative Math

Computation:

You're applying a known method. No creativity needed.

Computation vs. Creative Math

Computation:

You're applying a known method. No creativity needed.

Creative Math:

You're approaching a problem where the way to solve it isn't known. You have to *get creative*.

You can always play

You can always play

Not knowing what to do **is fun!**

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try.

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller
- try something deliberately unusual

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller
- try something deliberately unusual
- change the way data is represented

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller
- try something deliberately unusual
- change the way data is represented
- examine previous failures and why they failed

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller
- try something deliberately unusual
- change the way data is represented
- examine previous failures and why they failed
- play “what if”

You can always play

Not knowing what to do **is fun!**

Remember:

You can never really be **stuck on a problem**. There's always something more to do/try. Accept the discomfort of not knowing.

- make the problem smaller
- try something deliberately unusual
- change the way data is represented
- examine previous failures and why they failed
- play “what if”
- make a conjecture – then try to break it

Interlude

Activity:

Go to the board and represent the number 12 in different ways, ie. symbols, diagrams, pictures, arrangements, stories.

Promote creativity

Promote creativity

Make your work more creative by:

Promote creativity

Make your work more creative by:

- giving yourself space to work away from everything else

Promote creativity

Make your work more creative by:

- giving yourself space to work away from everything else
- giving yourself time to work – a beginning and an end

Promote creativity

Make your work more creative by:

- giving yourself space to work away from everything else
- giving yourself time to work – a beginning and an end
- leveraging the time as long as possible – don't cut thinking short

Promote creativity

Make your work more creative by:

- giving yourself space to work away from everything else
- giving yourself time to work – a beginning and an end
- leveraging the time as long as possible – don't cut thinking short
- being confident – during creation there's no such thing as being wrong

Promote creativity

Make your work more creative by:

- giving yourself space to work away from everything else
- giving yourself time to work – a beginning and an end
- leveraging the time as long as possible – don't cut thinking short
- being confident – during creation there's no such thing as being wrong
- be playful – tell jokes, try weird things, get silly

Inspirational video

Ministry of Silly Walks (link)

Interlude-revisited

Activity:

Go to the board and represent the number 12 in different ways, ie. symbols, diagrams, pictures, arrangements, stories.

Play with friends

Play with friends

Playing is more fun with good friends who promote / protect creativity!

Play with friends

Playing is more fun with good friends who promote / protect creativity!

- help develop suggestions: “Let’s see what happens with that.”

Play with friends

Playing is more fun with good friends who promote / protect creativity!

- help develop suggestions: “Let’s see what happens with that.”
- investigates errors precisely: “This example breaks things. What part needs changing?”

Play with friends

Playing is more fun with good friends who promote / protect creativity!

- help develop suggestions: "Let's see what happens with that."
- investigates errors precisely: "This example breaks things. What part needs changing?"
- makes room for uncertainty: "We can leave that question open for now."

Play with friends

Playing is more fun with good friends who promote / protect creativity!

- help develop suggestions: "Let's see what happens with that."
- investigates errors precisely: "This example breaks things. What part needs changing?"
- makes room for uncertainty: "We can leave that question open for now."
- shares control: "Which direction should we try to investigate next?"

Play with friends

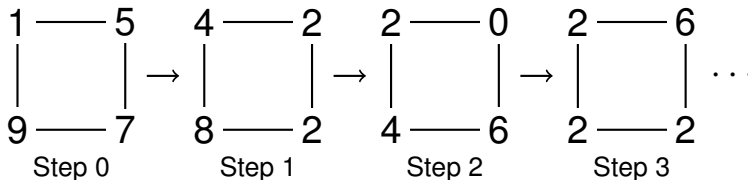
Playing is more fun with good friends who promote / protect creativity!

- help develop suggestions: "Let's see what happens with that."
- investigates errors precisely: "This example breaks things. What part needs changing?"
- makes room for uncertainty: "We can leave that question open for now."
- shares control: "Which direction should we try to investigate next?"
- protects the process of discovery: "would you like a hint, or more time to think?"

New problem

Activity:

Write four nonnegative integers around a square. Replace them simultaneously by the absolute differences between neighboring numbers. Keep repeating. What happens? How long does it take? Can you make something different happen?



Outline

- 1 SoPS Lecture 11
 - Konigsberg bridge problem
 - Math and creativity
 - Wrapping up

Question of the Day

Question of the Day

Puzzle

Take the numbers $\{1, 2, \dots, 2026\}$. What's the largest subset of them with the property that no number divides another number?

Final thoughts

Reflection

Final thoughts

Reflection

- Give an example of creativity in mathematics.

Final thoughts

Reflection

- Give an example of creativity in mathematics.
- Give an example of what you can do to help promote creativity.

Submit your responses by **Tonight!**